

Phase 1 - Week 3 - Lesson Day 1

Matplotlib Basics

Today's Objectives

Weeks 1 and 2 built the data handling foundation. Week 3 is about **seeing** that data. Matplotlib is the original, most widely used Python plotting library, and almost every other visualization tool (Seaborn, Pandas' own `.plot()`) is built on top of it. Today covers the four chart types you will use constantly: line, bar, histogram, and scatter.

- Build a line chart to show trends over a sequence
- Build a bar chart to compare categories
- Build a histogram to see the distribution of a numeric column
- Build a scatter plot to see the relationship between two variables

1. Line Chart

Definition: A line chart connects data points in order with a line - it is the standard choice for showing how something changes over a sequence, most commonly time.

```
import matplotlib.pyplot as plt

months = ["Jan", "Feb", "Mar", "Apr", "May"]
sales = [120, 150, 170, 140, 200]

plt.plot(months, sales)
plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.show()
```

2. Bar Chart

Definition: A bar chart uses rectangular bars to compare values across distinct categories - use this instead of a line chart when your x-axis is categorical (names, groups), not a continuous sequence.

```
cities = ["Mogadishu", "Hargeisa", "Kismayo"]
population = [2500, 800, 400]

plt.bar(cities, population)
plt.title("Population by City (thousands)")
plt.show()
```

3. Histogram

Definition: A histogram shows the *distribution* of a single numeric column - it groups values into "bins" (ranges) and shows how many values fall into each bin. This is the chart type you reach for to answer "what does the spread of this column look like?"

```
import numpy as np

scores = np.random.normal(70, 10, 200) # 200 simulated exam scores

plt.hist(scores, bins=15)
plt.title("Distribution of Exam Scores")
plt.xlabel("Score")
plt.ylabel("Frequency")
plt.show()
```

bins controls how many bars the range is split into - too few bins hides detail, too many makes the shape noisy. 10-20 bins is a reasonable starting point for most datasets.

4. Scatter Plot

Definition: A scatter plot places one dot per data point, using two numeric columns as the x and y position. This is the standard chart for seeing whether (and how) two variables relate to each other - the visual starting point before running any correlation or regression.

```
hours_studied = np.random.uniform(0, 10, 50)
exam_score = hours_studied * 6 + np.random.normal(0, 8, 50)

plt.scatter(hours_studied, exam_score)
plt.title("Hours Studied vs Exam Score")
plt.xlabel("Hours Studied")
plt.ylabel("Exam Score")
plt.show()
```

Chart	Use it when...
Line	Showing a trend over a sequence, usually time
Bar	Comparing values across distinct categories
Histogram	Showing the distribution/spread of one numeric column
Scatter	Showing the relationship between two numeric columns

Today's Exercise

Open **day-1-matplotlib.ipynb**. This notebook already contains fully solved answers for every exercise below, using generated sample data so it runs immediately without needing an uploaded dataset. Read through each

solved cell, run it yourself, and try changing the data/labels to make sure you understand what every line is doing - the goal is understanding the solution, not just re-typing it.

Exercise Questions (4 total - all solved in the notebook)

1. Line chart

Plot a line chart of 6 months of your own made-up revenue numbers, with a title and axis labels.

2. Bar chart

Plot a bar chart comparing 4-5 categories of your choice (e.g. product sales, test scores by subject).

3. Histogram

Generate 300 random values from a normal distribution and plot their histogram with 20 bins.

4. Scatter plot

Generate two related random numeric arrays (one influencing the other, like the worked example) and plot them as a scatter plot.

Once you and Halimo have reviewed and run all four solved exercises, move on to Lesson Day 2 (Seaborn & Plotly).